Test-Time Scaling for Visual Chain-of-Thought Models: Image-Input Diversity as the Key Driver

Date: 2026-04-14 **Status:** Runs 1 & 2 (pilot, n=30) + Run 3 (full hard_bench, n=292) complete

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1. Research Question

Can visual chain-of-thought (CoT) models benefit from test-time scaling (TTS) via majority voting? And if so, what is the source of candidate diversity that makes it work?

We compare two models that share the same 3B Qwen2.5-VL backbone but differ only in their reasoning strategy:

Model	Reasoning strategy	Parameters
Qwen2.5-VL-3B	Direct answer (no CoT)	3B
GRIT-3B	Visual CoT: think → grounded bbox → rethink → answer	3B (same backbone)

This pairing isolates the effect of visual CoT: same capacity, same weights, only the prompting and fine-tuning strategy differs.

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2. Experimental Setup

Benchmark

We evaluate on three vision tasks from hard_bench — chosen for hardness (< 60% for current 7B VLMs), recency (released late 2024), and vision-indispensability:

Task	Source	Format
VQA	MMMU-Pro (Sep 2024)	10-option MCQ, 30 academic disciplines
OCR	OCRBench v2 (NeurIPS 2024)	Free-form text extraction, 30 OCR types
Counting	MMStar instance-counting (NeurIPS 2024)	MCQ A/B/C/D

30 questions per task, per model, per run (90 per model per run). Questions were calibrated via a Pass-1 sweep (N=5, T=0.7) to fall in [20%, 70%] accuracy for Qwen3B.

TTS candidate recipe

Each question generates **9 candidate answers** varying two diversity axes — image augmentation and temperature:

#	Image augmentation	Temperature	#	Image augmentation	Temperature
0	Original	0.0 (greedy)	5	Edge enhance	0.7
1	Original	0.7	6	JPEG recompress	0.7
2	Paraphrase	0.7	7	Rotation 90	0.7
3	Brightness +30%	0.7	8	Paraphrase 2	0.7
4	Grayscale	0.7			

Final answer selected by majority vote (plurality) over the 9 candidates.

Two runs: isolating diversity sources

Run	Temperature	Image augmentations	Purpose
Run 1	Mixed (T=0.0 + T=0.7)	Yes	Full TTS pipeline
Run 2 (T=0 ablation)	All T=0.0	Yes	Isolate image-augmentation diversity

Run 2 is the critical experiment: by removing temperature stochasticity, we test whether image augmentations alone produce useful diversity.

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3. Results

3.1 Run 1 — Full TTS pipeline

	Qwen3B (direct)			**GRIT (visual CoT)**		
Task	Greedy	@9	Oracle@9	Greedy	@9	Oracle@9
VQA	30.0%	20.0%	53.3%	10.0%	13.3%	63.3%
OCR	3.3%	3.3%	6.7%	26.7%	30.0%	40.0%
Counting	36.7%	36.7%	93.3%	33.3%	30.0%	73.3%

Figure 5 — Greedy, majority @9, and oracle @9 for both models (30 questions/task). Two things stand out: (1) TTS gains are small or negative for both models, and (2) oracle@9 is dramatically higher than @9 — the correct answer is almost always *present* in 9 candidates but majority voting fails to select it (e.g. Qwen Counting: 93.3% oracle vs 36.7% vote).

3.2 The key experiment: Run 1 vs Run 2 (T=0 ablation)

Figure 5 — Run 1: Greedy vs Majority@9 vs Oracle@9
(9 candidates, mixed temperature + image augmentations, 30 questions/task)

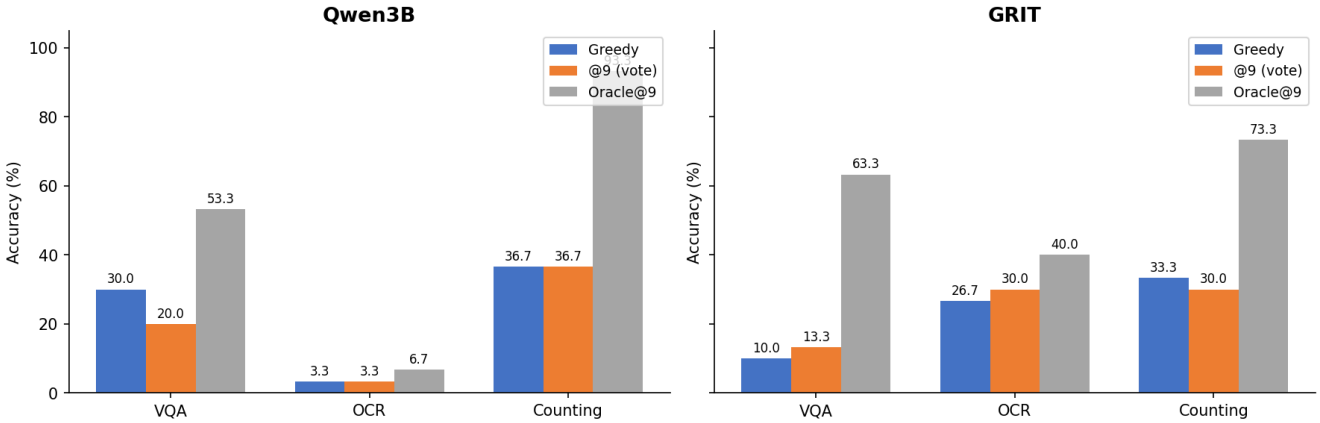
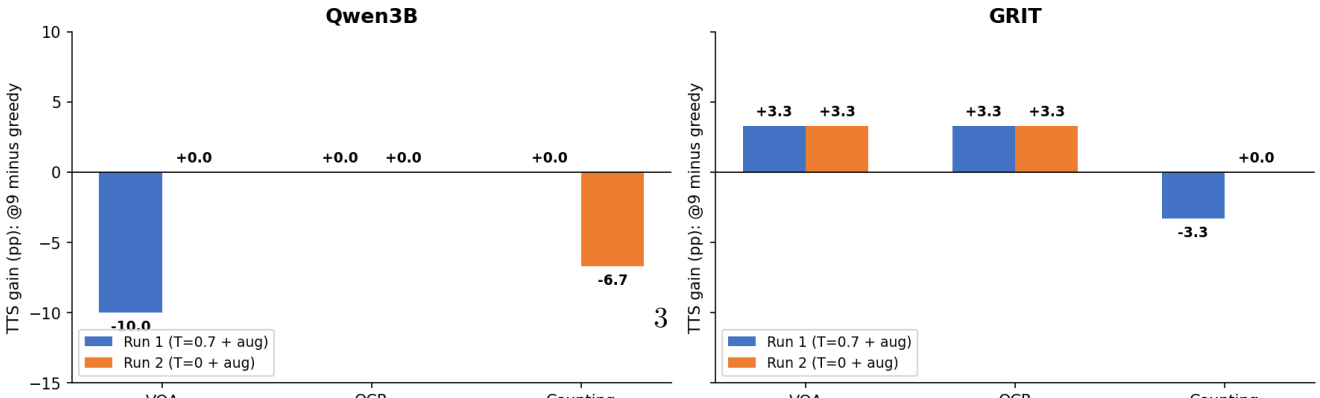


Figure 1: Figure 5 — Run 1: Greedy vs Majority@9 vs Oracle@9

Model	Task	Run 1 TTS gain	Run 2 TTS gain	What changed
Qwen3B	VQA	**-10.0pp**	0.0pp	Temperature hurt; removing it removes the harm
Qwen3B	OCR	0.0pp	0.0pp	No diversity source helps
Qwen3B	Counting	0.0pp	-6.7pp	Image augmentations add noise, no useful diversity
GRIT	**VQA**	**+3.3pp**	**+3.3pp**	**Same gain — temperature was irrelevant**
GRIT	**OCR**	**+3.3pp**	**+3.3pp**	**Same gain — temperature was irrelevant**
GRIT	Counting	-3.3pp	0.0pp	Removing temperature removes the harm

Figure 6 — TTS gain: Run 1 (mixed diversity) vs Run 2 (image-only diversity)
(positive = TTS helps, negative = TTS hurts; 30 questions/task)



4. Findings

Finding 1: Visual CoT models benefit from TTS via image-input diversity; direct-answer models do not

The T=0 ablation proves this cleanly. At T=0, Qwen produces the exact same answer regardless of image augmentation — brightness, grayscale, edge enhance, JPEG, rotation are all invisible to it. GRIT gains +3.3pp from those same augmentations because its grounding step (think → bbox → rethink) amplifies small visual perturbations into different reasoning chains and different final answers.

Finding 2: Temperature stochasticity hurts majority voting

For Qwen, temperature is the only source of diversity — and it generates wrong candidates that outvote correct ones (-10pp on VQA). For GRIT, temperature adds nothing beyond what image augmentations already provide. In both cases, removing temperature helps or is neutral.

Finding 3: Alternative voting strategies confirm Qwen does not benefit from TTS

The large oracle gap (oracle@9 » @9) raises the question: is plurality voting just a bad selector, and would a smarter strategy unlock the gains? We replayed the existing candidate data under five alternative voting rules:

Strategy	Rule
Greedy tiebreak	Plurality, but ties broken in favor of the greedy answer
Consistency filter (k=2,3)	Discard answers appearing fewer than k times, then vote
Supermajority (k=3,4,5)	Use greedy unless >= k candidates agree on an alternative

Qwen3B — best strategy is greedy itself. No voting rule improves over the greedy answer. The most conservative strategies (supermajority_5) simply recover the greedy baseline, confirming that TTS candidates add only noise for a direct-answer model.

Qwen3B	Greedy	Plurality	Best strategy	Oracle@9
VQA (n=100)	**27.0%**	20.0%	supermajority_5: 26.0%	48.0%
Counting (n=92)	39.1%	38.0%	supermajority_3: **42.4%**	91.3%

GRIT — voting strategies yield small gains over greedy. Unlike Qwen, GRIT benefits from candidate aggregation regardless of the specific voting rule, with supermajority_3 reaching +10pp over greedy on VQA (Run 2).

GRIT	Greedy	Plurality	Best strategy	Oracle@9
VQA (n=30, Run 2)	10.0%	13.3%	supermajority_3: **20.0%**	56.7%
OCR (n=30, Run 1)	13.3%	16.7%	plurality: **16.7%**	20.0%

The pattern is consistent: for Qwen, the optimal strategy converges to "just trust the greedy answer" — TTS adds nothing. For GRIT, multiple voting strategies improve over greedy, confirming that the visual CoT candidates carry useful signal that aggregation can exploit. However, all strategies remain far below the oracle, indicating that answer-level voting alone cannot close the gap — token-level confidence signals or learned verifiers would be needed.

Finding 4: Stochasticity does not predict TTS effectiveness

We initially expected CoT models to be more stochastic, making TTS more effective. A calibrated entropy study (30 questions, 10 draws each at T=0.7) showed the opposite: GRIT is *less* stochastic than Qwen on all three tasks (entropy delta: -0.23 to -0.47 bits). Yet GRIT benefits from TTS while Qwen does not. What matters is not how much diversity a model produces, but whether that diversity is *structural* (grounding-driven) or *random* (temperature-driven).

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5. Scaled validation (n=292) — redefining TTS as pass@9

The pilot results raised one key question: with only 30 questions per task and a persistent oracle gap, is the TTS effect simply noise, or is there real signal that a better selector could exploit? We scaled to the full **hard_bench (n=292)** across all four configurations (Qwen3B × GRIT × standard × T=0 ablation).

Metric redefinition. At scale, no answer-level voting strategy beats greedy for either model (see §5.5). Rather than continue to chase an optimal voter, we fix the TTS metric to **pass@9** — a question counts as correct under TTS if *any* of the 9 candidates is correct. This is the upper bound a perfect selector could reach, and it isolates the question we actually care about: *does candidate diversity contain the right answer?* Closing the gap between pass@9 and a deployable selector is left as a limitation.

5.1 Headline: TTS lifts accuracy sharply on both models

Figure A — TTS gain (pass@9 minus greedy) on the full 292-question benchmark. Every configuration shows a substantial lift; the **standard recipe (T=0.7 + image augmentations)** yields the largest gains, reaching **+30.5pp for GRIT** and +24.3pp for Qwen3B overall. Even the pure image-augmentation ablation (T=0) produces +16.8pp (Qwen) and +19.9pp (GRIT) — confirming at scale the pilot finding that image diversity alone is a real driver.

5.2 Per-task effect and model comparison

Figure B — Per-task breakdown under the standard recipe. The two tasks where models have headroom (VQA, Counting) also show the largest TTS gains (+29–43pp). OCR gains are modest on both models (+5–7pp), consistent with the pilot: OCR is a fine-grained exact-match task where augmentations produce fewer *correctly different* candidates.

Figure D — Visual-CoT (GRIT) vs direct (Qwen3B) under the standard recipe. Baseline accuracy is nearly identical (29.5 vs 29.8% overall), but **GRIT gains more from TTS** on every task, widening the gap to 59.9 vs 54.1% under pass@9. The visual-grounding step gives GRIT access to *structurally different* candidates that direct decoding cannot produce.

5.3 Temperature amplifies TTS

Figure C — At scale, temperature sampling roughly doubles the TTS gain for both models (Qwen: +16.8 → +24.3pp; GRIT: +19.9 → +30.5pp overall). This refines Finding 2 from the pilot: temperature

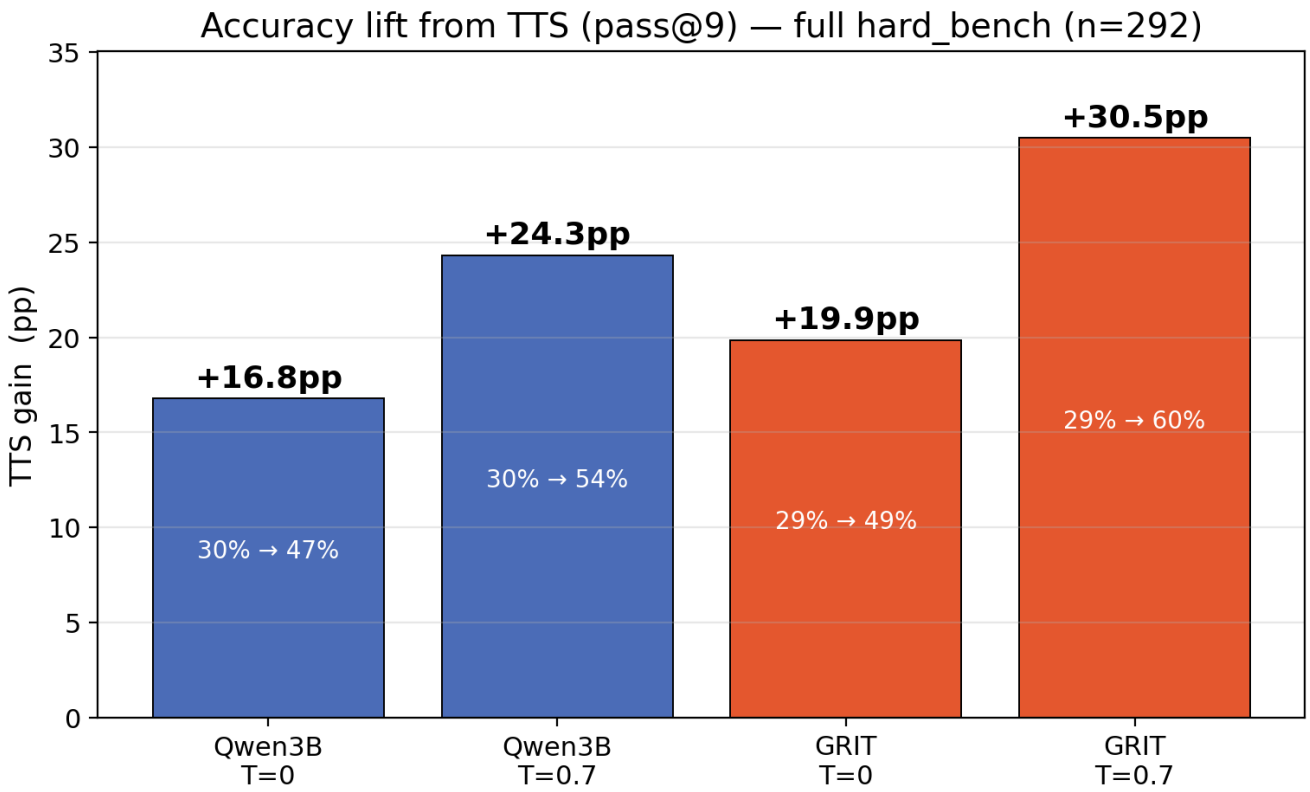

Figure 3: Figure A — Accuracy lift from TTS (pass@9) across 4 configs

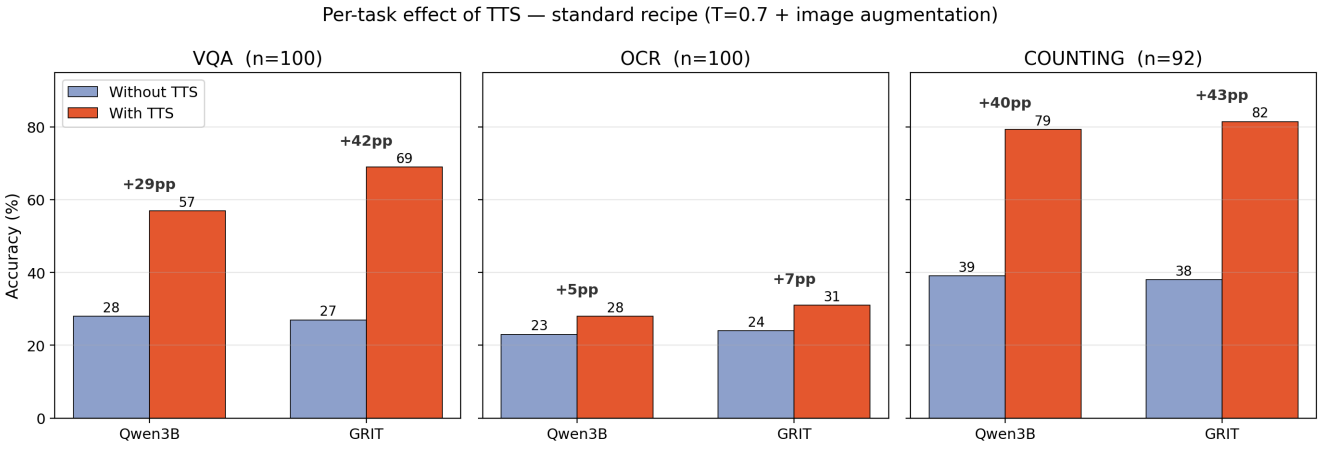

Figure 4: Figure B — Per-task effect of TTS, standard recipe

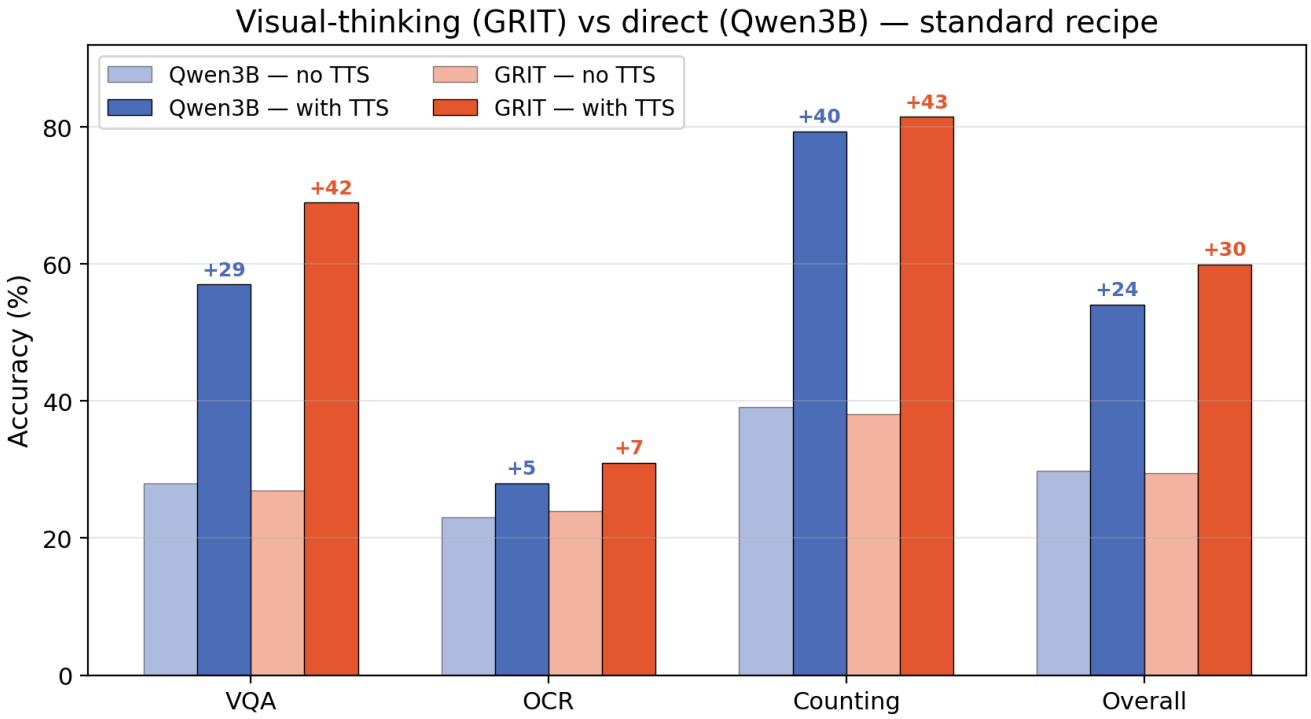

Figure 5: Figure D — GRIT vs Qwen3B head-to-head

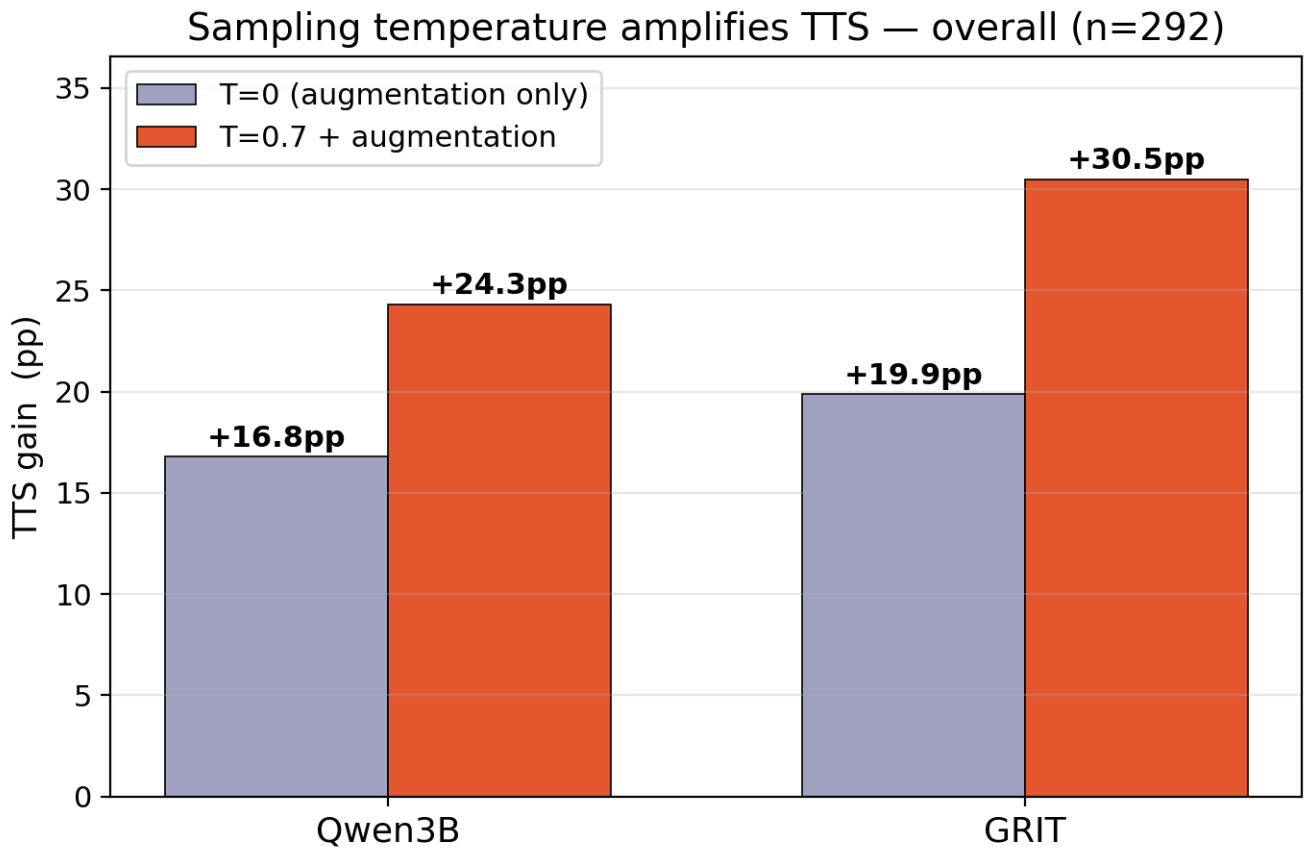

Figure 6: Figure C — Temperature amplifies TTS

does generate useful diversity once we measure in the pass@9 regime — the earlier "temperature hurts" result was specific to plurality voting, where wrong candidates can outvote correct ones. With a perfect selector, both diversity sources compound.

5.4 Which augmentation drives GRIT's diversity?

To understand where GRIT's candidate diversity comes from, we conditioned on questions where the original image (T=0) failed and measured how many of those failures each augmented candidate uniquely rescues — i.e., it is the *only* augmented candidate that gets the question right.

Augmentation	GRIT rescue%	Qwen rescue%	GRIT unique	Qwen unique
Rotation 90°	12.3%	12.6%	**10**	4
Edge enhance + paraphrase	**13.2%**	13.0%	4	1
JPEG recompress	10.8%	12.1%	5	3
Brightness/contrast	11.3%	12.1%	3	4
Edge enhance	11.8%	11.6%	1	0
Paraphrase (text only)	10.4%	**14.4%**	1	1
Grayscale	10.8%	13.5%	0	3

rescue% = fraction of original-failed questions this slot answers correctly; unique = questions where this slot is the only correct augmented candidate (T=0 ablation, n=292)

The **raw rescue rates are nearly identical across all augmentations and across both models** (11–14%), meaning each augmentation recovers a similar number of failures in absolute terms. The critical difference lies in *which* questions they recover. For Qwen, augmented candidates largely rescue the **same questions** — when one augmentation works, most others do too. For GRIT, **rotation_90 uniquely rescues 10 questions that no other augmentation touches**, and each augmentation type unlocks a distinct subset of failures.

This is consistent with GRIT's visual grounding mechanism. A 90° rotation fundamentally changes the spatial layout of the image: objects that were side-by-side become vertically stacked. GRIT's grounding step predicts bounding boxes before answering, so a rotated image forces it to attend to a different spatial region, producing a qualitatively different reasoning chain. Brightness shifts and JPEG compression alter pixel values but not spatial structure, so the grounding step is minimally affected — and for Qwen, which has no grounding step, rotation is just another perturbation with no mechanism to translate spatial change into different reasoning.

Figure F — Stacked bars showing how much oracle coverage comes from original-image candidates alone vs what augmented candidates uniquely add (T=0 ablation). GRIT gains more from augmentation than Qwen both at T=0 and T=0.7, with the gap widest under temperature sampling where GRIT's grounding step compounds spatial and stochastic diversity.

5.5 Answer-level voting does not close the oracle gap

At n=292 we replayed six voting strategies (plurality, greedy-tiebreak, greedy-unless-supermajority, consistency filter, logprob-sum, logprob-mean). For both models and across all tasks, **no strategy outperforms the greedy baseline**. Options-logprob voting (restricted to the A–D/A–J subset where options are well-defined) is no better. This is why we adopt pass@9 as the TTS metric: the signal is in the candidate pool, but extracting it requires more than answer-level aggregation.

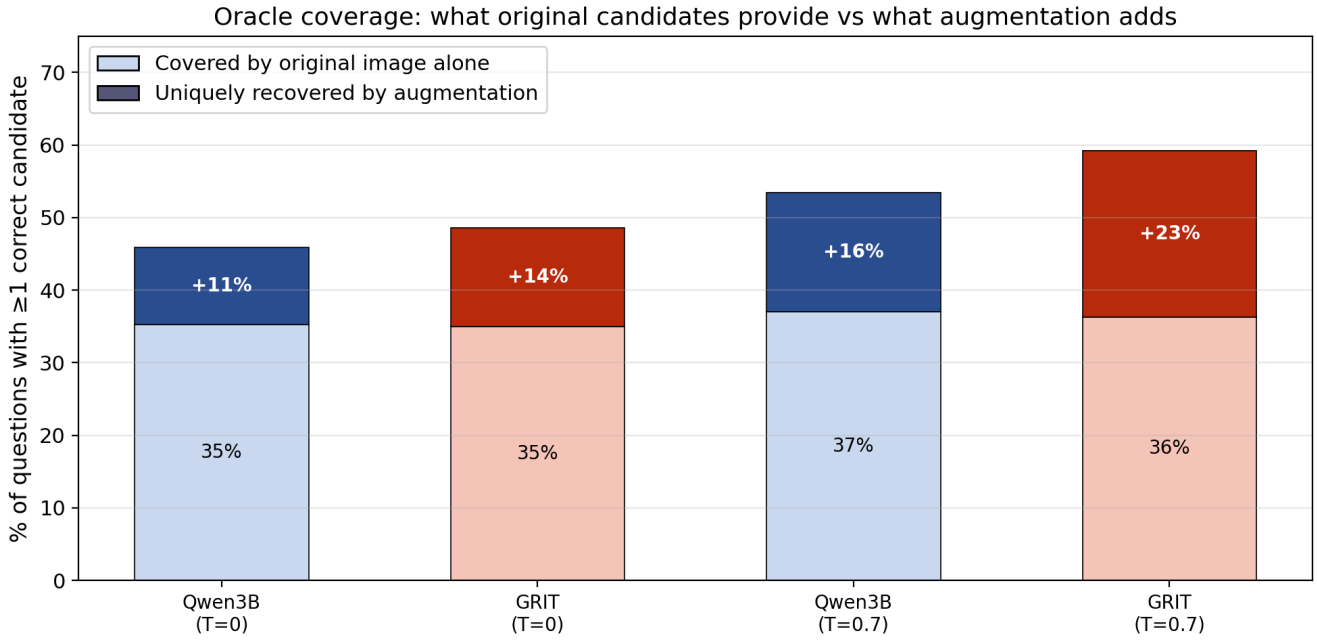

Figure 7: Figure F — Oracle coverage: original vs augmented candidates

Full strategy-by-strategy tables are in `results/analysis/scale_results.json`; we treat selector design as an open problem rather than a contribution of this work.

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6. Limitations

- **TTS metric = pass@9 (oracle upper bound).** Due to time constraints we did not identify a voting strategy that matches pass@9; answer-level voting actually under-performs greedy at scale. Reported TTS gains therefore represent what a perfect selector could achieve, not deployable accuracy.
- **Model pair:** Only Qwen3B vs GRIT-3B tested. DeepEyesV2-7B (agentic CoT) was piloted on 3 questions; full evaluation is planned.
- **OCR evaluation** remains conservative due to strict exact-match scoring; edit-distance re-scoring is pending.

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7. Next Steps

1. **Selector design to close the pass@9 → deployable gap.** Logprob-weighted voting on counting/MCQ subsets and learned lightweight verifiers are the natural next experiments.
2. **DeepEyesV2 full evaluation** — test whether the image-diversity finding extends to agentic CoT.
3. **OCR re-scoring** with character-level edit distance to separate model error from evaluation artifact.